EXPLORE AND DISCOVER

EXPLORATION

MAIN DESTINATIONS

Low Earth orbit – LEO

- ESA will maximise use of the International Space Station up until its expected end of use in 2030.
- Sophie Adenot and Raphaël Liégois will undertake the next two long-duration missions.
- ESA will continue development of the LEO Cargo Return Service with a hard deadline of 2029 for the first flight.
- ESA will procure limited existing ISS commercial cargo services, with European industry involvement, to ensure Europe's short term utilisation of the ISS.
- ESA will investigate options for additional flight opportunities for ESA astronauts, maintaining access to on-orbit science and technology utilisation, supporting the drive towards lunar and Martian exploration.

Moon

- ESA will fulfil its part in the Artemis programme:
  1. delivering European Service Modules 5 and 6.
  2. continuing work on the lunar Gateway, developing communication, habitation, refuelling capabilities while being mindful of evolutions in the timeline and partnership;
- ESA will develop payloads to demonstrate early-on mobility, robotics and survivability means on the lunar surface;
- ESA will deliver the first Argonaut lander – carrying a surface mobility asset to the Moon along with the navigation package NovaMOON in 2030;
- ESA will accelerate European exploration of the Moon by flying small missions - new fast-tracked opportunities with moderate investment;
- ESA will enable Europe's very first commercial lunar communication service with the Lunar Pathfinder launch.

Mars

- ESA will guarantee a continuous flow of ground-breaking scientific discoveries and seamless data relay support with Trace Gas Orbiter;
- The Rosalind Franklin mission is on course to be launched to Mars in 2028. It will demonstrate Europe's capability of entry, descent, and landing and enable a pioneering quest to uncover signs of possible life beneath the Martian surface;
- ESA will study a re-oriented use of the Earth Return Orbiter, originally designed for Mars Sample Return, as a one-way exploration science mission to Mars.

EXPLORATION SCIENCE

Exploration Science makes use of the full spectrum of platforms and destinations (ground-based facilities, analogue environments, sub-orbital and orbital platforms, and Moon

Exploration delivers value in science, economics, international cooperation and inspiration. For example, the current programme has 1600 active scientists from 36 countries, sustains over 15 000 high-quality jobs, contributes €2.8 billion in gross added value to the European economy, and reaches 30 000 teachers and 1.3 million students annually.

and Mars), to access the unique conditions and vantage points that enable discoveries impossible to achieve on Earth alone.

Exploration Science has two streams:

- Exploration-focused science: advancing the fundamental knowledge and capabilities required for successful exploration by reducing risk and maturing critical technologies in key domains: local resources; space environment and effects; crew health and performance; habitation.
- Exploration-enabled science: Exploration also drives curiosity-driven, world-class research across the HRE destinations and platforms, encompassing fundamental and applied life and physical sciences, planetary geology, and the search for life beyond Earth.

FUTURE PREPARATION

For future human and robotic missions, ESA will study the mission options and orchestrate technology derisking to close critical capability gaps and accelerate the early maturation of breakthrough solutions to deliver top notch and still cost-effective future missions:

- Advanced guidance navigation and control, increasing the reliability of next-generation landers;
- Cryogenic throttleable engines, preparing the evolution of current landers towards heavy landing alternatives;
- Launch-abort technologies enabling the development of European end-to-end crewed flight solutions and preparing Europe for the post-ISS low Earth orbit landscape;
- Autonomous mobility solutions compatible with the hostile Moon and Mars environments;
- Advanced thermal control and power systems enabling the lunar night survivability and operability;
- Radioisotope power systems, opening the path to long duration surface missions.

EXPLORATION BACKGROUND AND RATIONALES

From Ulf Merbold, the first ESA astronaut to fly in space, who worked on Spacelab during its inaugural flight in late 1983, to Sophie Adenot, who will be the next ESA astronaut to fly to the International Space Station, Europeans have excelled in human spaceflight.

ESA was a founder of the ISS partnership; its contributions include the Columbus laboratory, Nodes 2 and 3, the Automated Transfer Vehicle spacecraft plus astronauts' favourite ISS location: the Cupola. This largest window in space is employed for operating the robotic Canadarm2, monitoring spacecraft traffic and looking down to Earth 600 km below.

Building on its significant achievements so far, in human and robotic exploration Europe is a highly regarded contributing partner in LEO and is assuming more ambitious roles in international exploration campaigns to the Moon and Mars.

This growth in ambition should lead to a realistic but sustained increase in financial means – subject to political decision – commensurate with Europe's stance among world-leading players and the growing number of private actors and competitors in space exploration.

Space exploration is accelerating as both established and emerging players are pushing boundaries, driven by bold political agendas. Europe is at crossroads, with decades-long partnerships being questioned. With the global competition heating up, Europe must act decisively to valorise past investments and ongoing activities, making sure Europeans keep benefitting from it.

At CM25, Human Robotic and Exploration is seeking approval to push forward with delivering missions that will start implementing the Explore2040 strategy and step up autonomy for Europe to access exploration destinations, leaving options open for new cooperation. ESA will also present a flexible programme to accelerate the preparation of future projects ensuring Europe's role in space exploration for the decades to come, while also realising the benefits of its work in a volatile international funding environment.